

CEL: Comprehensive Counterfactual Explanations Library and Benchmark

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What is a counterfactual explanation?

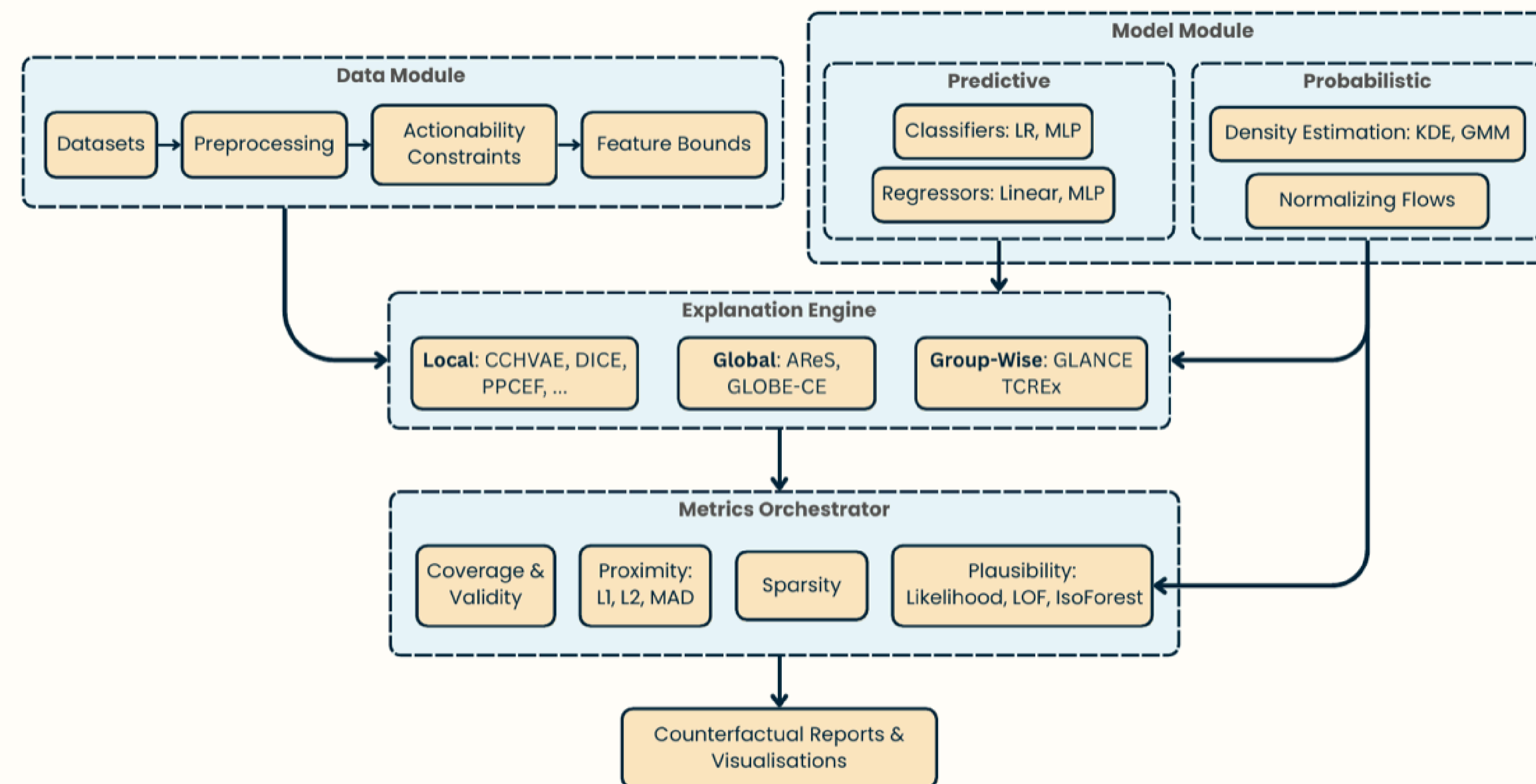
A small input change that flips a model's prediction.

LOAN APPLICATION EXAMPLE

Feature	Original	Counterfactual
Age	35 years	35 years
Employment	Full-time	Full-time
Credit history	6 years	6 years
Loan amount	€15,000	€15,000
Monthly income	€3,000	€3,500
Monthly debt	€1,200	€700

Prediction **Declined** **Approved**

Only 2 of 6 features change.



18 datasets

Adult Census · Audit · Bank Marketing · Blobs · Credit Default · Digits · German Credit · Give Me Some Credit · HELOC · Law · Lending Club · Moons · Wine · Synthetic · Concrete · Diabetes · Yacht · SCM20D

14 methods

Local
Wachter · Artelt · DiCE · CCHVAE · PPCEF · CEM · CEGP · CADEX · SACE · CEARM

Global
ARes · GLOBE-CE

Group-wise
GLANCE · T-CREX

2 backbones / task

Classification

Multi-Layer Perceptrons (MLP) · Logistic Regression

Continuous targets

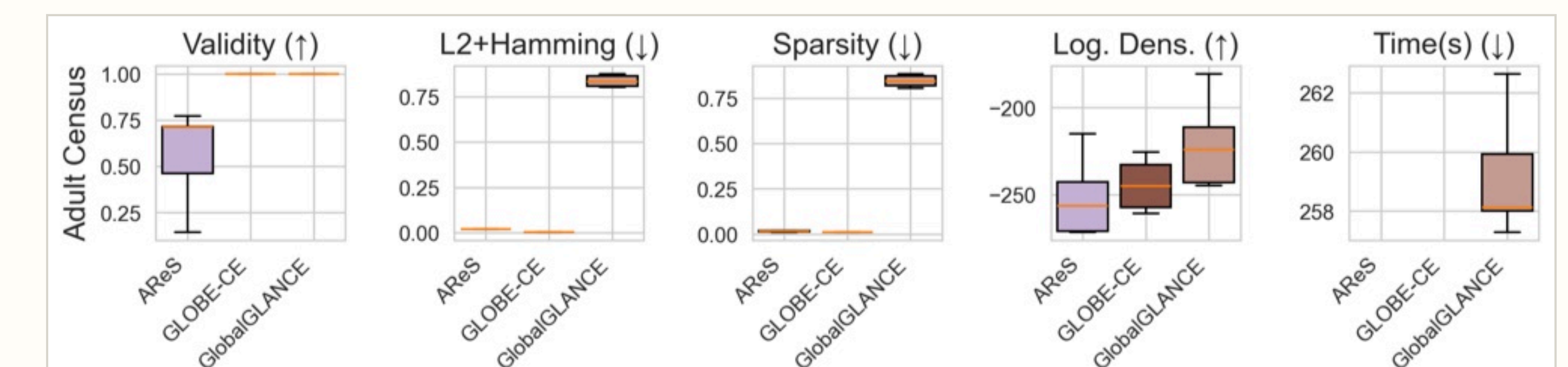
Linear Regression · MLP Regressor

9 classification metrics

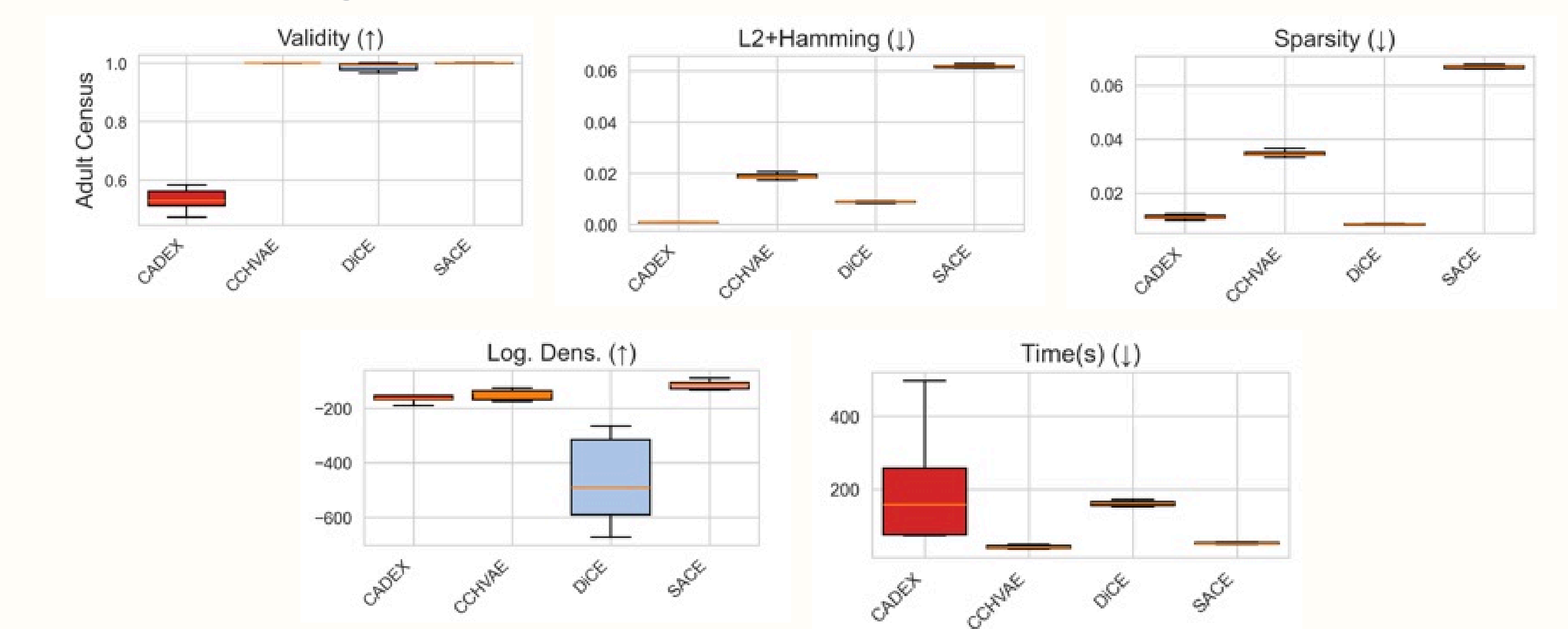
Coverage · Validity · Sparsity · Prob. plausibility · Log-density · LOF · Isolation Forest · Proximity (L2 / L2-Hamming) · Runtime

Results

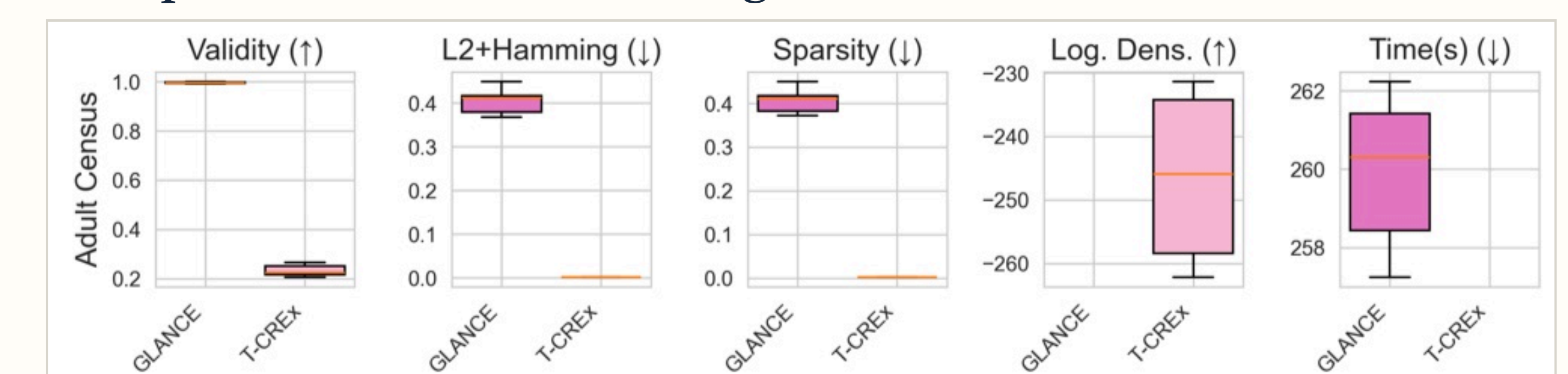
Global: validity and failures



Local: no single winner



Group-wise: reach versus change



Three contributions

CONTROL

Controlled evaluation protocol

COVER

Broad benchmark across CE paradigms

EXTEND

Extensible open-source benchmark workbench



Code & project